SOFTWARE ENGINEERING PROJECT

Prometheus:
Computer Vision-Based Gym Machine Occupancy Detection and Spatial Analytics

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Chapter 1

Introduction

1.1 Background

The fitness industry has seen a significant surge in membership, yet the technological infrastructure within gyms remains largely stagnant. Most facilities still rely on basic membership check-ins, leaving a massive data gap between the front desk and the gym floor. For members, the lack of real-time information regarding equipment availability leads to inefficient workouts and "availability uncertainty." For owners, the absence of granular data on how floor space is utilized results in suboptimal equipment purchasing and layout decisions. With the advancement of Computer Vision and the ubiquity of CCTV systems, there is a significant opportunity to transform passive video feeds into actionable spatial intelligence.

1.2 Problem Statement

Current gym environments operate as "black boxes" once a member passes the front desk. This lack of transparency creates two critical points of failure:

1. Dynamic Bottlenecks: Without real-time visibility, members experience "availability uncertainty." This leads to routine disruption and peak-hour congestion that management cannot see or redirect, directly impacting member retention.

2. Static Asset Management: Owners currently make high-stakes capital decisions—such as purchasing five new treadmills or removing a power rack—based on anecdotal evidence rather than objective occupancy data. This leads to "dead zones" where expensive square footage is

underutilized, and "over-taxed zones" where equipment wear-and-tear is accelerated.

3. Infrastructure Scalability: Traditional IoT solutions (individual machine sensors) are cost-prohibitive and fragile in high-impact environments. There is currently no scalable way to capture floor-wide spatial intelligence using existing infrastructure.

1.3 Solution Overview

Prometheus solves these issues by implementing a non-intrusive AI system that utilizes existing gym CCTV cameras. The system uses an Edge-to-Cloud architecture where local processing units analyze video streams to identify machine states (Free vs. Occupied). This metadata is then synchronized to a cloud database, powering a real-time mobile tracker for members and a spatial analytics dashboard for management.

Prometheus bridges the gym's "data gap" by transforming existing CCTV infrastructure into a real-time spatial intelligence network. Using a privacy-first Edge-to-Cloud architecture, the system processes video feeds locally to detect equipment occupancy without storing sensitive biometric data [1]..

The solution delivers value through two integrated touchpoints:

1. For Management: A centralized analytics dashboard that maps high-traffic "heat zones" and equipment duty cycles, enabling data-driven decisions on floor-plan optimization and capital expenditure.

2. For Members: A real-time mobile interface that eliminates "availability uncertainty" by showing live equipment status.

1.3.1 Features

1. **Real-time Occupancy Tracker:** A live mobile interface showing which machines are currently in use or available to help members plan their sets.

2. **Spatial Heatmaps:** A web-based visual representation of high-traffic areas and underutilized equipment for facility owners.

3. **Progress Logger:** A feature for members to record sets, reps, and weights, integrated with visual progress charts and historical data.

4. **Workout Routine Generator:** An intelligent routine creator tailored to user goals, physical profile, and real-time equipment availability.

5. **Automated Utilization Reports:** Weekly data summaries for owners regarding peak hours, machine popularity, and equipment ROI.

1.4 Target User

The Prometheus system is designed for two distinct demographics, with a primary focus on facility management and business intelligence:

- **Gym Owners and Facility Managers (Primary Users):** These are the central stakeholders who utilize the system to drive operational efficiency. They require high-level spatial analytics to identify equipment usage patterns, minimize "dead zones," and optimize facility layouts. By leveraging the web-based dashboard, these users transform raw computer vision data into actionable business insights to improve equipment Return on Investment (ROI) and long-term member retention.

- **Gym-goers (Secondary Users):** Individual patrons who benefit from the downstream data generated by the system. These users range from beginners to advanced athletes who utilize the mobile interface to minimize wait times and follow AI-generated routines. While they are frequent interactors with the platform, their experience is an extension of the primary analytical framework established for the facility.

1.5 Terminology

- **Computer Vision (CV):** A field of Artificial Intelligence that enables computers to derive meaningful information from digital images or videos.
- **YOLO (You Only Look Once):** A state-of-the-art, real-time object detection system used by Prometheus for identifying equipment and patrons [2].
- **Edge Computing:** Processing data near the source (the camera) to reduce latency and enhance privacy by keeping raw video footage local.
- **Region of Interest (ROI):** Specifically defined coordinates within a video frame that correspond to individual pieces of gym equipment.
- **Occupancy Logic:** The algorithmic determination of a "busy" state based on the spatial intersection of "Person" and "Machine" detection boxes.

Chapter 2

Literature Review and Related Work

2.1 Competitor Analysis

The competitive landscape for fitness technology is currently divided between manual tracking applications and high-cost industrial hardware. Prometheus positions itself as an automated, non-intrusive solution that leverages existing infrastructure to provide high-utility data at a lower entry cost for facility owners.

The following table provides a detailed comparison between Prometheus and existing solutions in the market, highlighting the gaps in real-time spatial intelligence.

Feature	**Prometheus**	**Manual Loggers**	**IoT Sensors**	**Gym Dashboards**
Real-time Occupancy	Automated CV on RTSP streams	Manual inspection or self-report	Entry/exit counts only	Manual check-in required
Installation Cost	Low; reuses CCTV infrastructure	Zero; pen-and-paper logs	High; per-station sensors	Moderate; software and kiosks
Member Utility	High; live heatmap and forecasts	High for workout logging only	Low; not member-facing	Moderate; schedules only
Spatial Analytics	Full heatmaps and utilisation trends	None	Counts only; no mapping	Check-in data only
Privacy	Edge-only; no raw video shared	High; no video collected	High; no biometric data	Moderate; third-party SaaS
Scalability	Per-camera; add feeds easily	Poor; effort grows linearly	Hardware-limited scaling	Software tier-limited
Maintenance	Periodic retraining; CCTV upkeep	Staff training only	Battery, firmware, hardware failures	Vendor-managed; kiosk servicing

Table 2.1: Comparison Table of Competitive Fitness Solutions

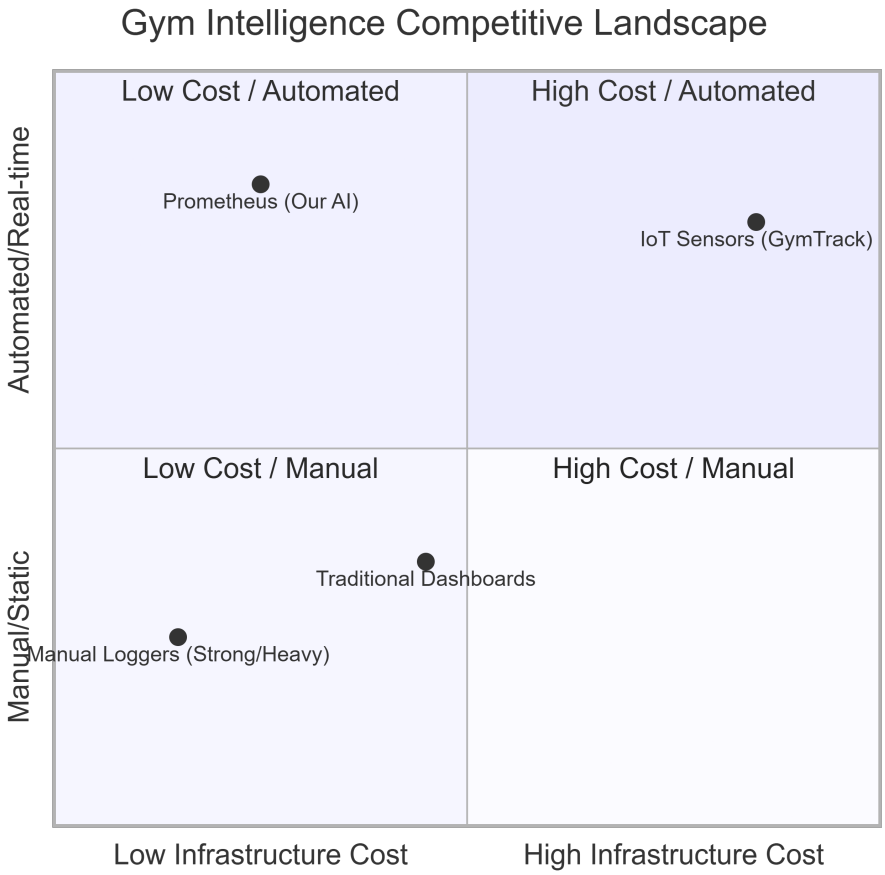


Figure 2.1: Competitive Landscape: Automation vs. Infrastructure Cost

2.2 Literature Review

To build a technically sound and feasible system, this project draws from several key academic works and technical advancements in the fields of Computer Vision and Distributed Systems.

2.2.1 Real-Time Object Detection (YOLO Architecture)

The foundation of the occupancy detection engine is based on the "You Only Look Once" (YOLO) architecture. Research by Redmon et al. established that frame-rate inference is possible by treating detection as a single regression problem. This project specifically evaluates the use of [2] and YOLOv10 [3]. The selection of YOLOv10 informs our choice

for edge deployment, as its elimination of Non-Maximum Suppression (NMS) significantly reduces computational latency on local hardware.

2.2.2 Vision-Based Gym Monitoring (GymCam)

The "GymCam" project by Khurana et al. demonstrated the viability of using unconstrained camera feeds to track multiple users and exercises simultaneously. Their research indicates that while fine-grained action recognition is complex, identifying machine occupancy through spatial overlap is highly reliable. Prometheus extends this logic by utilizing a multi-camera RTSP network to provide floor-wide spatial analytics.

2.2.3 Large-Scale Multi-View Datasets (M3GYM)

For model training, this project utilizes the M3GYM dataset [4]. Unlike standard datasets, M3GYM provides multi-view labels in real-world gym environments characterized by high occlusion. This informs the design of our "Occupancy Logic," allowing the system to distinguish between a person walking near equipment and a person actively utilizing a machine.

2.2.4 Privacy-Preserving Edge Computing

A primary constraint for gym-based AI is member privacy (GDPR/PDPA [1]). Current literature suggests that performing inference at the "edge" allows raw video data to be discarded immediately after metadata extraction. This informs our choice of an Edge Processing Layer (e.g., ASUS NUC), ensuring that no identifiable video frames are ever transmitted to the cloud.

2.2.5 Retail and Commercial Space People Counting

Vision-based people counting has been extensively deployed in retail environments to measure footfall and dwell time at product shelves. Systems such as those described by Albiol et al. use overhead camera feeds combined with background subtraction and blob tracking to count

entrants and exiters at store thresholds. These systems share Prometheus's architectural pattern of fixed-camera inference on a constrained zone, but operate in single-plane entry contexts rather than complex multi-zone floor plans. The retail domain's emphasis on conversion-rate analytics (dwell time versus purchase rate) directly parallels Prometheus's per-equipment utilisation metrics, demonstrating that spatial intelligence derived from CCTV feeds has measurable commercial value beyond security.

2.2.6 Smart Building and Office Occupancy Detection

Occupancy detection for energy management in smart buildings is a well-studied problem. Hailemariam et al. showed that computer-vision-based room occupancy outperforms PIR motion sensors in accuracy because vision captures head-count rather than mere movement. Building Management Systems (BMS) from vendors such as Siemens Enlighted and Cisco Spaces combine ceiling-mounted depth sensors with Wi-Fi probe requests to estimate desk and meeting-room occupancy at scale. Prometheus adopts the same principle of inferring occupancy from non-contact sensing, extending it from single-room granularity to per-equipment granularity across a large open floor plan.

2.2.7 Crowd Density Estimation in Public Spaces

Crowd density estimation research, originating from public safety applications in transportation hubs and stadiums, provides techniques directly applicable to high-density gym floors. Zhang et al.'s CSRNet demonstrated that dilated convolutional networks can generate density maps from heavily occluded scenes without per-person bounding boxes, making inference tractable even when individuals overlap. While Prometheus targets lower crowd densities than stadium scenarios, the density-map paradigm informs our heatmap visualisation layer, where pixel-level density estimates are aggregated into zone-level utilisation scores displayed on the facility dashboard.

2.2.8 Sports and Athlete Tracking Systems

Professional sports analytics systems such as Second Spectrum (NBA/EPL) and Hawk-Eye (tennis, cricket) demonstrate that multi-camera triangulation can yield sub-centimetre positional accuracy for tracking individuals in a bounded arena. At a lower cost tier, research by Gade et al. using overhead thermal cameras in sports halls achieved robust person tracking under varying lighting without privacy concerns, since thermal imagery is non-identifiable. These approaches validate the principle of camera-network-based spatial tracking used by Prometheus, while also establishing the thermal camera as a privacy-safe sensing modality that may be considered for future iterations where GDPR compliance is critical.

2.2.9 Hospital and Healthcare Facility Flow Analysis

Patient flow and staff utilisation tracking in hospitals presents challenges structurally similar to gym equipment monitoring: high asset density, frequent short-duration interactions, and strong privacy requirements. Systems reviewed by Ajami et al. use RFID badges combined with zone antennas to track patient dwell times at care stations. Computer-vision alternatives, explored by Bao et al., replace RFID with overhead camera pose estimation to infer staff activity without wearable hardware. The healthcare domain's regulatory focus on anonymisation (HIPAA/PDPA) aligns with Prometheus's edge-processing privacy model and reinforces the design decision to transmit only aggregated metadata rather than video or biometric identifiers.

Chapter 3

Requirement Analysis

3.1 Stakeholder Analysis

The Prometheus project involves multiple parties that influence or are affected by the system's deployment. Identifying these stakeholders ensures that both the technical AI requirements and the business-end objectives are met.

- **Gym Owners and Facility Managers (Primary Users):** Business operators who require data-driven insights to optimize equipment placement and facility layout. They utilize the web dashboard for spatial analytics and ROI assessments.
- **Gym Members (Secondary Users):** Individual athletes and gym patrons who utilize the facility. They are the primary consumers of the real-time occupancy data and progress tracking features. Their main interest is minimizing workout friction and maximizing time efficiency.
- **Product Owner:** The individual responsible for the strategic vision of the Prometheus ecosystem. They ensure that the system remains compliant with data privacy regulations (PDPA/GDPR [1]) and that the AI capabilities align with the intended problem domain.

3.2 User Stories and Use Cases

3.2.1 User Stories

The following user stories define the core functional requirements from the perspective of the key stakeholders:

- **Gym Member:** "As a gym member, I want to see real-time machine occupancy via my mobile app so that I can plan my sets without waiting for equipment."
- **Gym Member:** "As a gym member, I want to receive an AI-generated workout routine based on currently available machines so that I can have a continuous, uninterrupted workout session."
- **Gym Owner:** "As a gym owner, I want to view a spatial heatmap of my facility so that I can identify 'dead zones' and relocate underutilized equipment to maximize my floor space value."
- **Gym Owner:** "As a gym owner, I want to access historical utilization reports so that I can make informed decisions about purchasing new equipment based on actual demand."
- **Product Owner:** "As a product owner, I want to ensure all raw video data is processed locally at the edge so that the system maintains member privacy and remains compliant with regional data protection laws."

3.2.2 Use Cases

The following use cases detail the interaction between the users and the Prometheus system.

Use Case 1: View and Update Real-Time Machine Status (Fully-Dressed)

Primary Actor: Gym Member
Supporting Actor: Edge Processing Unit (System Actor)
Goal: To allow members to identify available equipment in real-time and provide accurate live updates via the mobile application.
Stakeholders and Interests:

- **Gym Member:** Needs low-latency, accurate updates to avoid "availability uncertainty" and plan their workout efficiently.

- **Gym Owner:** Requires a reliable system that encourages member retention by reducing training friction.
- **Product Owner:** Interested in ensuring data transparency and privacy through real-time anonymization and compliance with data protection standards.

Preconditions:

- The facility has at least 5 HD IP cameras active and streaming via RTSP to the local edge hardware.
- The Computer Vision (CV) model is loaded and initialized on local edge-processing hardware.
- The member has the Prometheus iOS application open and connected to the service.

Main Success Scenario:

1. The Edge Processing Unit receives a live video frame from the gym floor.
2. The CV system identifies machine occupancy states within the facility.
3. The system validates the "Occupied" or "Free" state using internal occupancy logic.
4. The Edge Unit extracts anonymized metadata (coordinates and activity labels) and immediately discards the raw video to protect member privacy.
5. The extracted metadata is transmitted to the cloud-based data management system.
6. The member opens the Prometheus iOS application to view the live status of the facility.
7. The application displays a visual map where equipment icons are color-coded: Green for "Free" and Red for "Occupied".
8. The system triggers a real-time update to all active iOS clients the moment a machine state changes.

Extensions (Alternative Flows):

- **4a. Occlusion Handling:** If the model momentarily loses visual track of a person, temporal smoothing logic prevents a false "Free" status trigger.
- **7a. Notification Subscription:** If a member has subscribed to a specific exercise, the app sends a push notification the moment the required machine becomes vacant.
- **8a. Privacy Restriction:** Members are provided with machine status indicators only and cannot view live CCTV feeds directly for security and privacy reasons.

Use Case 2: Analyze Facility Utilization Trends (Fully-Dressed)

Primary Actor: Gym Owner
Goal: To identify peak usage times and equipment popularity to optimize business operations and capital expenditure.

Stakeholders and Interests:

- **Gym Owner:** Needs data-backed insights to justify equipment investments (CapEx) and floor plan optimization.
- **Facility Manager:** Seeks to optimize staffing, cleaning schedules, and operational efficiency based on traffic volume.
- **Gym Member:** Indirectly benefits from an improved gym layout and a more frictionless training environment.

Preconditions:

- Historical utilization data has been tracked and stored in the database over a period of time.
- The owner has authorized access to the Prometheus Web Dashboard.

Main Success Scenario:

1. The Gym Owner logs into the PC-based web dashboard.

2. The owner selects a date range for utilization analysis (e.g., daily, weekly, or monthly).
3. The system retrieves the "time-in-use" records for every machine from the historical database.
4. The dashboard generates visual heatmaps of the facility to identify hotspots and "dead zones".
5. The dashboard provides reports on the most popular equipment and peak usage times.
6. The owner analyzes the visual intelligence to identify overcrowded areas or underutilized assets.
7. The owner makes informed decisions regarding facility layout optimization or future capital expenditures.

Extensions (Alternative Flows):

- **4a. Operational Optimization:** The owner identifies traffic trends to adjust staffing or cleaning schedules based on real-time traffic volume.
- **7a. CapEx De-Risking:** The owner decides to remove underused assets identified in "dead zones" to make space for higher-demand equipment.

3.3 Use Case Diagrams

The Prometheus system utilizes a dual-interface approach to serve its distinct user groups. The following diagrams and descriptions outline the functional requirements for both Gym Members and Gym Owners, as well as the automated interactions performed by the system itself.

3.3.1 Gym Member Use Cases

The member-facing use cases focus on personal workout efficiency, progress tracking, and data privacy.

- **UC-1: View Real-Time Machine Status:** Allows members to view a live map of the gym floor where equipment is color-coded based on availability (Green for Free, Red for Occupied).
- **UC-2: Subscribe to Vacancy Notifications:** Users can select specific machines or exercises; the system sends a push notification the moment the required equipment becomes vacant.
- **UC-3: Log Workout Progress:** A standard interface for logging weights, sets, and reps, which are then visualized using progress charts.
- **UC-4: Workout Routine(Rule Based):** Uses a structured logic to create a personalized plan based on the user's physical profile and goals.
- **UC-5: Manage Privacy & Data Rights:** Provides members with the ability to exercise their rights under the PDPA [1], such as requesting a copy of their data or the deletion of their personal logs.
- **UC-6: Analyze Machine Traffic:** Provides a real-time overview of equipment usage states across the facility floor.

3.3.2 Gym Owner Use Cases

The owner-facing use cases prioritize spatial intelligence, facility optimization, and business analytics.

- **UC-7: View Spatial Heatmaps:** Generates visual heatmaps to identify high-traffic hotspots and underutilized "dead zones".
- **UC-8: Generate Utilization Reports:** Accesses a historical database to produce daily, weekly, or monthly reports on equipment popularity and peak usage times.
- **UC-9: Optimize Facility Layout:** Enables owners to make data-backed decisions on relocating assets to improve floor flow and utilization.

- **UC-10: Manage Operational Schedules:** Helps owners optimize staffing, cleaning, and maintenance schedules based on real-time and historical traffic volume.

3.3.3 System Use Case

This automated use case represents the core AI functionality that enables the user-facing features.

- **UC-12: Detect and Update Machine State:** An automated process where the Edge Processing Unit analyzes CCTV streams, identifies occupancy via intersection logic, and transmits anonymized metadata to the cloud while discarding raw video frames.

3.4 User Interface Design

3.4.1 Web Application

The Prometheus Web App features a Camera Management dashboard with a live camera grid displaying AI-detected occupancy using bounding boxes. A status panel monitors edge processing performance and latency. The interface also includes spatial overlays for heatmaps and camera controls for easy zone calibration and equipment labeling.

3.4.2 Mobile Application

The Prometheus Mobile App provides real-time gym occupancy through a live capacity meter and a searchable equipment list with availability and wait times. Users can join a waitlist and receive notifications when machines become available, with filters for quick browsing by category.

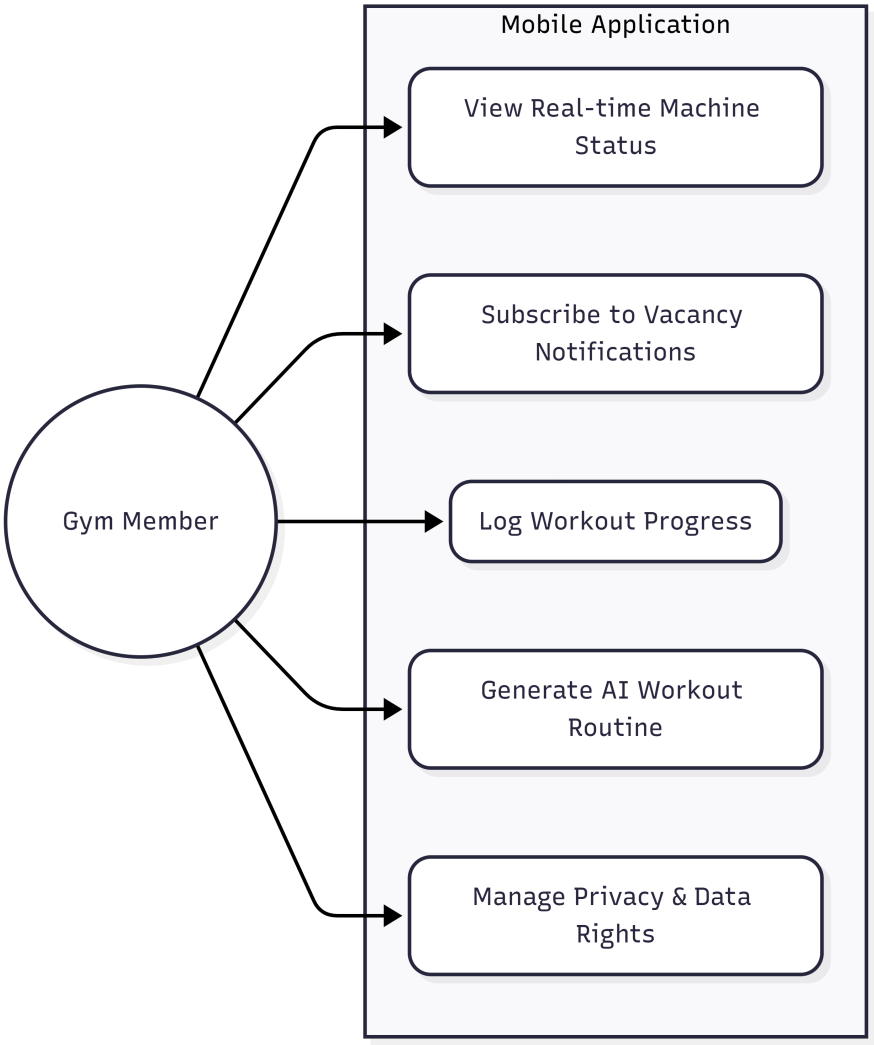


Figure 3.1: Use Case Diagram for Gym Members

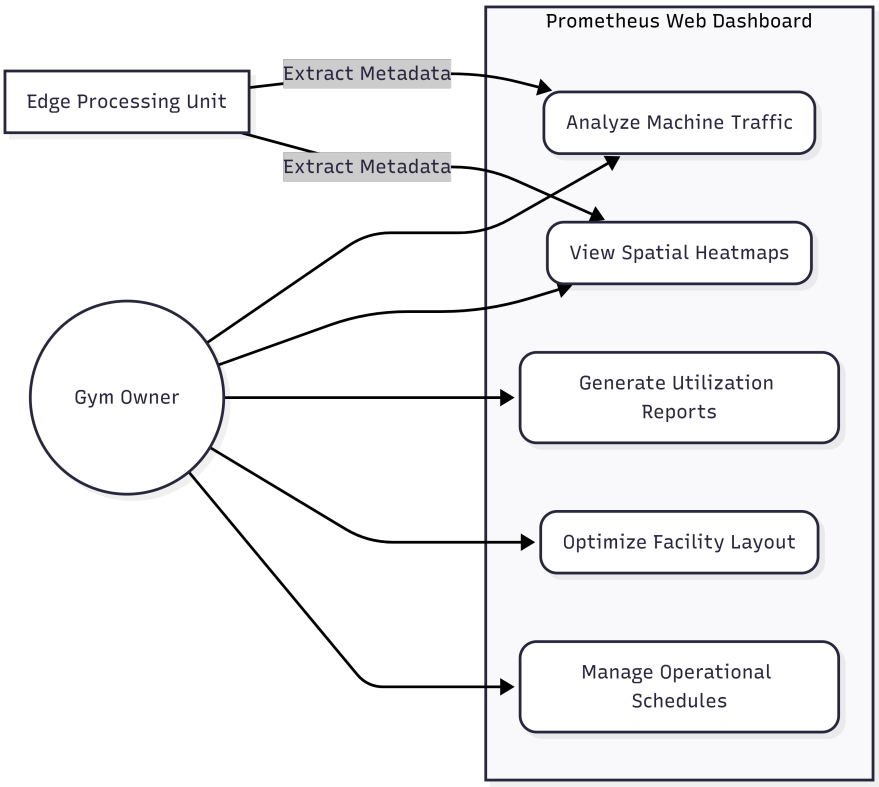


Figure 3.2: Use Case Diagram for Gym Owners

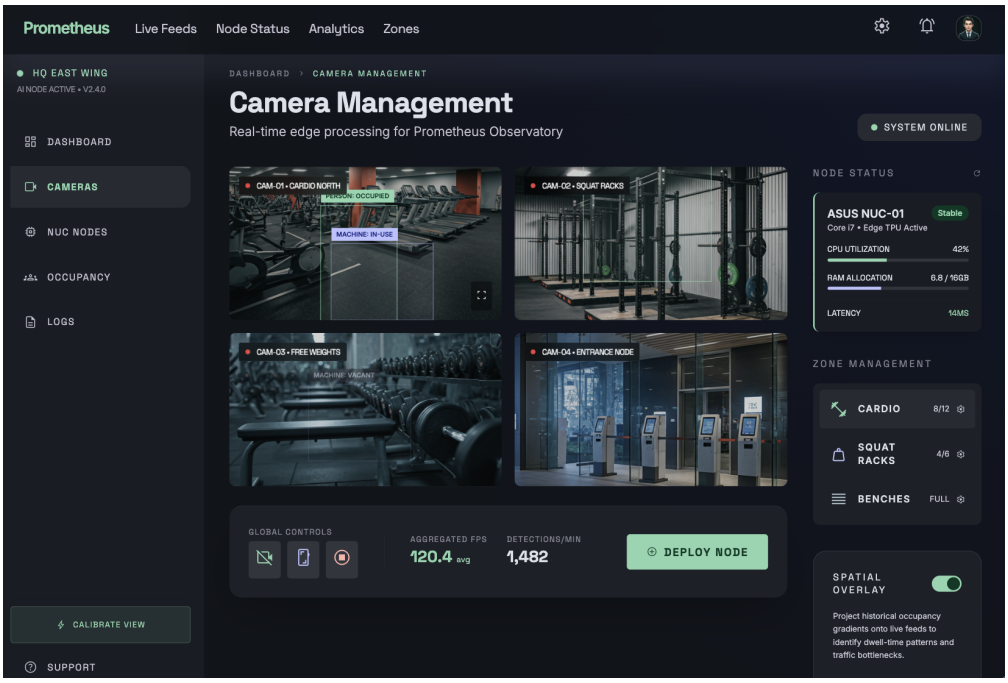
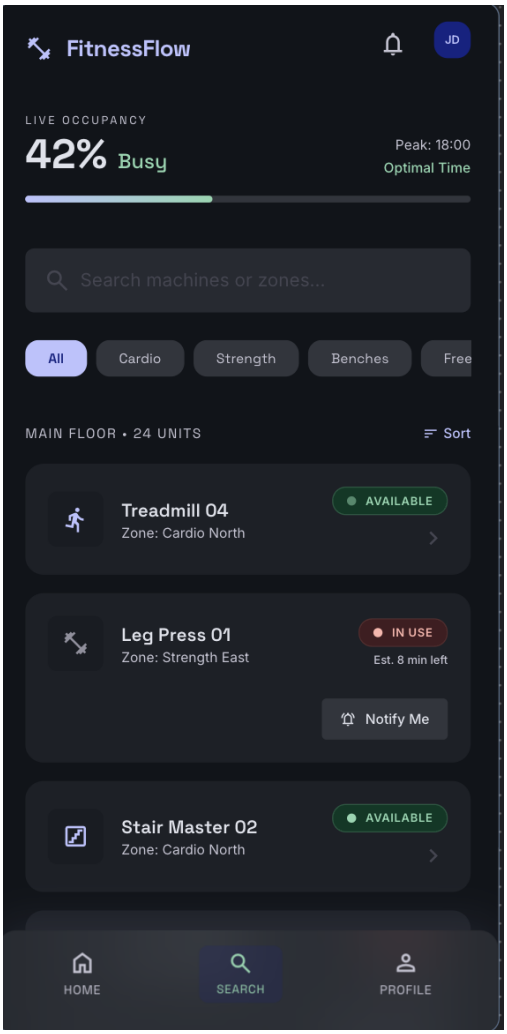


Figure 3.3: Prometheus Web Application Interface (Owner Dashboard)

(a) Overview



(b) Waitlist and Details

Figure 3.4: Prometheus Mobile Application Interface

Chapter 4

Software Architecture Design

4.1 System Architecture Overview

The Prometheus system follows a distributed Edge-to-Cloud architecture designed to balance real-time performance with data scalability. The architecture is composed of four primary layers: the Data Ingestion Layer (IP Cameras), the Edge Processing Layer (Inference Engine), the Cloud Service Layer (Data Management), and the Presentation Layer (Client Applications).

As illustrated in Figure 4.1, the process begins at the edge to ensure low latency and privacy. Metadata is then pushed to the cloud to serve the analytical needs of gym owners and the real-time needs of members.

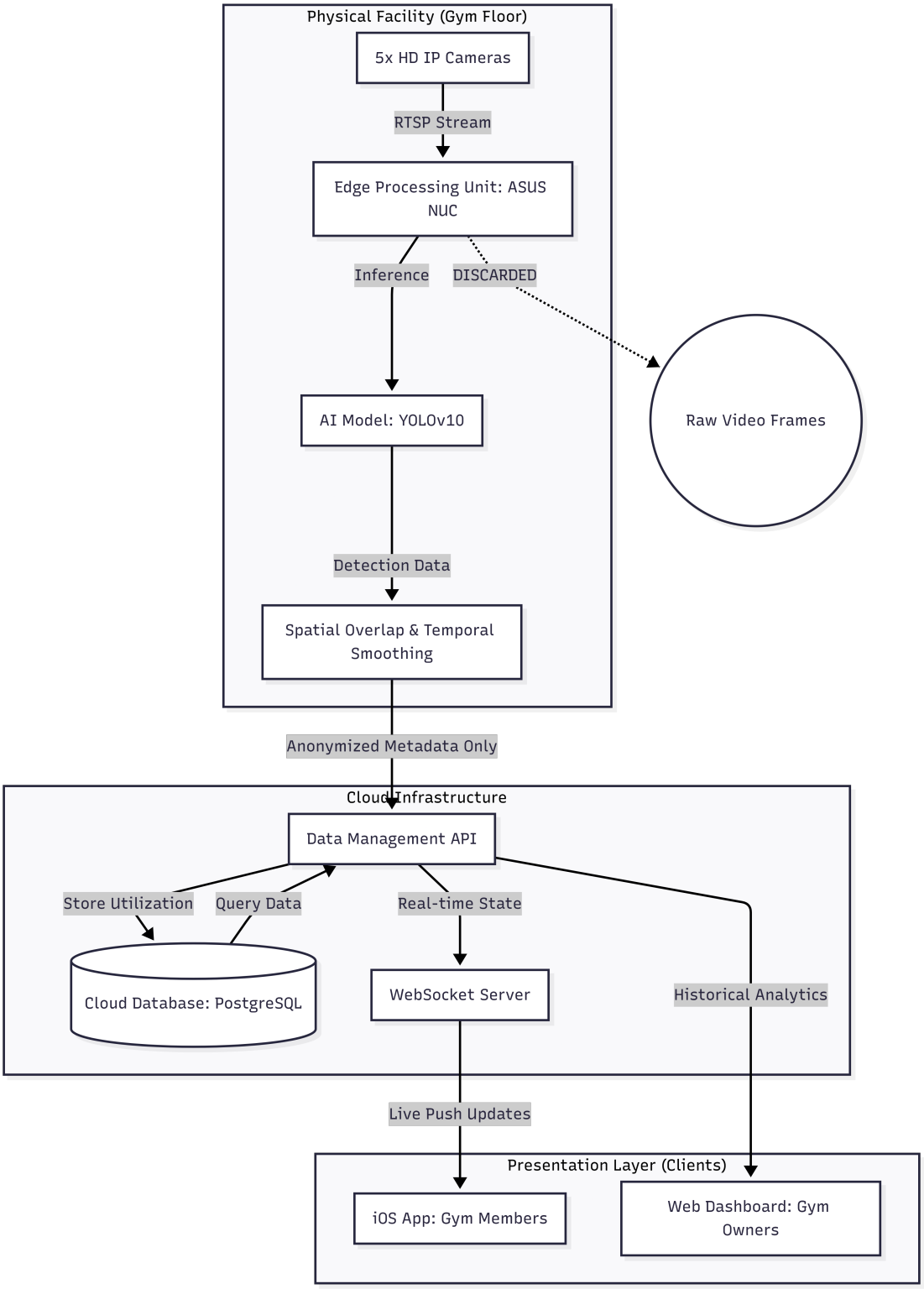


Figure 4.1: Prometheus Distributed Edge-to-Cloud System Architecture

The system begins with 5 HD IP cameras streaming high-definition video via RTSP to a local Edge Processing Unit (e.g., ASUS NUC). This unit performs real-time object detection and state classification, immediately discarding raw video frames to ensure member privacy. The resulting anonymized metadata is transmitted to a PostgreSQL-backed [5] cloud database, which serves as the central repository for historical utilization data. Finally, the Presentation Layer delivers live availability updates to the iOS application and spatial heatmaps to the PC Web Dashboard.

4.2 Software Design

The runtime behavior of Prometheus is driven by a continuous inference loop. When a member interacts with a machine, the system must detect, validate, and broadcast this state change within seconds.

Primary Sequence: Occupancy Update

1. The Edge Engine retrieves a frame and identifies "Person" and "Machine" objects using the YOLO model.

2. The system applies a **Spatial Overlap Algorithm** to determine if a person is actively using a machine. This logic calculates the intersection of the person's bounding box with a pre-defined Region of Interest (ROI) for that specific machine.

3. To prevent "flickering" from momentary occlusions or passersby, the system implements a **Temporal Smoothing** buffer: a state is only changed to "Occupied" if the overlap persists for at least 5 consecutive seconds.

4. Once validated, the status is pushed to the cloud, triggering a WebSocket event that updates the machine's color on all active iOS client maps.

4.3 AI and Data Design

4.3.1 Data Sources and Collection

The Prometheus AI models are developed using a hybrid data strategy to ensure both general robustness and facility-specific accuracy:

- **Public Benchmarks:** We utilize the M3GYM [4] and QEVD datasets for baseline training. These sources provide millions of frames of traditional weightlifting and machine-based exercises from multiple CCTV-style angles.
- **Self-Collected Data:** To fine-tune the model for the specific layout of local facilities (such as the KEFF gym), we collect approximately 2,000 frames of facility-specific equipment.
- **Data Cleaning:** All training data undergoes perspective correction and manual annotation of "Free" vs. "Occupied" states. For privacy compliance, all human faces in the training set are blurred before being fed into the training pipeline [1].

4.3.2 Model Design

The core of the Prometheus intelligence is based on the **YOLO (You Only Look Once)** architecture [2]. We have selected **YOLOv10** [3] as the primary model for deployment on the ASUS NUC edge hardware.

- **Reasoning:** YOLOv10 was chosen over traditional R-CNN models because it offers the highest inference speed (frames per second) while maintaining the precision necessary for occupancy detection. Furthermore, YOLOv10's architecture eliminates Non-Maximum Suppression (NMS) overhead, which is critical for running multiple camera streams on a single edge device.
- **Integration:** The model is integrated into the system via an ONNX or TensorRT execution provider to maximize the utilization of the edge unit's GPU/NPU.

4.3.3 Evaluation

To ensure the system meets the established success criteria, the AI components are evaluated against the following metrics:

- **Mean Average Precision (mAP):** To measure the model's accuracy in identifying machines and people.
- **State Accuracy:** The percentage of correctly identified "Free" vs. "Occupied" states, with a target of $\geq 95\%$.
- **End-to-End Latency:** The time elapsed from a person sitting on a machine to the iOS app reflecting the "Occupied" status. Target: < 5 seconds.
- **False Positive Rate:** The frequency at which the system incorrectly marks a machine as occupied due to environmental noise or passersby.

Chapter 5

Software Development

5.1 Schedule

The Prometheus project is structured into a 16-week development lifecycle. The schedule is divided into five distinct operational sections that progress from foundational infrastructure to final system evaluation, ensuring that the Computer Vision (CV) pipeline is validated before full integration with the user interfaces.

The timeline prioritizes the development of the core AI models and the Edge-to-Cloud metadata pipeline during the first half of the project. This is followed by parallel development of the iOS mobile application for members and the analytical Web Dashboard for facility owners. The final weeks are dedicated to rigorous system-wide evaluation to verify that the CV accuracy meets the established success criteria.

Figure 5.1: Prometheus Project Gantt Chart: 16-Week Development Timeline

5.1.1 Phase Descriptions

Based on the visual timeline presented in the Gantt chart, the project phases are defined as follows:

- **April 2026: Phase 1 — Infrastructure Setup, Data Acquisition, and Prototyping:** This initial phase focuses on the physical installation of the CCTV hardware and site surveys at the facility. Key tasks include data collection and annotation, establishing basic human-to-machine classification models, and developing early prototypes.
- **January 2027: Phase 2 — AI Model and Backend Pipeline Development:** Resuming after a scheduled break, this phase involves rigorous data preprocessing and the training of the occupancy detection models. The focus is on model validation, optimization, and the establishment of the core CV-to-Database pipeline to handle metadata transmission.
- **February 2027: Phase 3 — Mobile Application and Web Dashboard Implementation:** This phase centers on the development of the user interfaces for both gym members and owners. Work includes building the iOS application (incorporating the machine tracker and notification system) and the web-based dashboard for spatial heatmaps, followed by comprehensive UI/UX refinement and testing.
- **March 2027: Phase 4 — System Integration and Final Evaluation:** The concluding phase involves system-wide API wiring and integration to ensure all components communicate seamlessly. Efforts are dedicated to accuracy and performance testing against real-world scenarios, culminating in the production of the final evaluation report and project submission.

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